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MICROSOFT AZ-104 AZURE ADMINISTRATOR COMPLETE LEARNING HANDBOOK

Module-wise visual guide based on Microsoft Learn: Cloud Shell, ARM/Bicep, Identity, Governance, Networking, Storage,

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1 WELCOME - READ THIS FIRST

This handbook follows the AZ-104 learning flow you shared. Each page explains one domain in simple language with diagrams, checklists, and real admin steps. Use it for study, interview revision, and practical Azure labs.

- Learn the concept
- See the step-by-step flow
- Practice the lab in Azure

2 WHAT THIS HANDBOOK COVERS

- Azure Cloud Shell and admin prerequisites
- JSON ARM templates, parameters and outputs
- Microsoft Entra ID, users, groups and licensing
- RBAC, SSPR, Azure Policy and initiatives
- Virtual networks, NSG, DNS, peering and routing
- Azure Load Balancer, App Gateway and Network Watcher
- Storage accounts, Blob, Files, SAS, endpoints and redundancy
- VMs, availability, VMSS, App Service and containers
- Azure Monitor, Log Analytics, alerts, backup and recovery

3 HOW TO STUDY THIS BOOK

- Read one page at a time and complete the matching Microsoft Learn module.
- Do every exercise in a lab subscription or sandbox.
- Create your own notes for commands, portal paths and mistakes.
- Use the final checklist before exam or interview.

4 TABLE OF CONTENTS

5 AZURE ADMINISTRATOR LEARNING JOURNEY

Understand Cloud & Identity → Govern Access & Policy → Build Storage & Compute → Connect Networking → Monitor Backup & Recover → Pass AZ-104

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AZ-104 PREREQUISITES, AZURE CLOUD SHELL & ARM TEMPLATES

Start with the admin tools, shell environment, and infrastructure-as-code foundation.

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1 AZURE ADMIN PREREQUISITES

- Know basic networking: IP, DNS, routing, firewalls.
- Understand servers, virtualization, and operating systems.
- Practice Azure Portal, Azure CLI, Azure PowerShell and Cloud Shell.
- Understand Microsoft Entra ID and RBAC basics.
- Use a lab subscription or Microsoft Learn sandbox when available.

2 WHAT IS AZURE CLOUD SHELL?

Azure Cloud Shell is a browser-based command-line environment for managing Azure. It gives you Bash or PowerShell, common Azure tools, and authenticated access from the portal.

Open Portal → Start Cloud Shell → Choose Bash/PowerShell → Run commands

3 HOW CLOUD SHELL WORKS

- Runs in a Microsoft-managed container.
- Uses your signed-in Azure identity for access.
- Can use a storage account for persistent files.
- Good for quick admin tasks, scripts, and learning.
- Not designed for long-running production jobs.

4 WHEN TO USE CLOUD SHELL

- Quickly check resources without local tools.
- Run Azure CLI or PowerShell commands from any device.
- Test examples from Microsoft Learn exercises.
- Troubleshoot access, deployments and resource states.
- Use local tools for heavy automation or long sessions.

5 DEPLOY AZURE INFRASTRUCTURE BY USING JSON ARM TEMPLATES

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ARM templates are JSON files that describe Azure resources in a repeatable way. They help admins deploy consistent infrastructure.

```
{
 "$schema": "...",
 "parameters": { },
 "variables": { },
 "resources": [ ],
 "outputs": { }
}
```

Explore template structure → Create template → Add parameters → Deploy → Review outputs

Parameters make templates flexible. Outputs return values after deployment. Use Bicep for cleaner authoring when possible.

6 LAB FLOW - CREATE AND DEPLOY AN ARM TEMPLATE

1. Create resource group → 2. Write JSON template → 3. Add parameters → 4. Deploy with CLI/Portal → 5. Validate resource

Exam focus: understand template sections, parameters, outputs, deployment flow, and when ARM/Bicep helps with repeatable infrastructure.

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DESCRIBE THE CORE ARCHITECTURAL COMPONENTS OF AZURE

Understand Azure accounts, physical infrastructure, and management hierarchy.

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1 WHAT IS MICROSOFT AZURE?

Microsoft Azure is a cloud platform for building, running, securing and managing applications and infrastructure. Azure provides compute, storage, networking, identity, security, monitoring, AI, databases and many other services on a global cloud infrastructure.

Account → Subscription → Resource Group → Resource → Monitor

2 AZURE ACCOUNTS & SUBSCRIPTIONS

- Azure account is your sign-in identity.
- Subscription is a billing, access and resource boundary.
- One tenant can have many subscriptions.
- Use separate subscriptions for prod, dev, test or business units.
- Apply RBAC and Azure Policy at the right scope.

3 PHYSICAL INFRASTRUCTURE

- Geography: data residency area such as India, US, Europe.
- Region: datacenter location where resources run.
- Availability zone: physically separate datacenter location inside a region.
- Region pair: paired region for resilience and recovery scenarios.

4 MANAGEMENT INFRASTRUCTURE - SCOPE AND INHERITANCE

Management Group
Organize subscriptions and apply governance at scale

↓

Subscription
Billing and access boundary

↓

Resource Group
Logical container for related resources

↓

Resources
VMs, VNets, storage accounts, databases

Important idea: access, policy and locks can be inherited from parent scopes. Assign at the highest scope only when you really need broad coverage.

5 MANAGEMENT BASICS

- Use tags for cost and ownership.
- Use locks to prevent accidental deletion.
- Use Azure Advisor for recommendations.

6 ADMIN CHECKPOINT

- Can you explain tenant vs subscription?
- Can you choose a region and availability option?
- Can you place resources in correct resource groups?

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UNDERSTAND MICROSOFT ENTRA ID FOR AZURE ADMINISTRATORS

Directory, cloud identity, AD DS comparison, P1/P2 and Entra Domain Services.

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1 EXAMINE MICROSOFT ENTRA ID

- Cloud identity and access management service.
- Stores users, groups, devices, app registrations and service principals.
- Provides authentication, authorization and app access.
- Tenant is a dedicated instance of Entra ID.

2 ENTRA ID VS ACTIVE DIRECTORY DS

Feature	Entra ID	AD DS
Location	Cloud	On-prem / VM
Protocol	OAuth, SAML, OIDC	Kerberos, LDAP, NTLM
Main use	Cloud apps, Azure access	Domain join, GPO, legacy apps
Admin focus	Identity, MFA, RBAC	Domain controllers, OUs, GPO

3 ENTRA ID AS DIRECTORY SERVICE FOR CLOUD

- Central identity for Microsoft 365, Azure and SaaS apps.
- Supports SSO for enterprise applications.
- Controls access with Conditional Access and MFA.
- Supports app registration, service principals and managed identities.

4 COMPARE ENTRA ID FREE, P1 AND P2

Plan	Typical features
Free	Basic user and group management
P1	SSPR, Conditional Access, dynamic groups, hybrid identity
P2	Identity Protection, Privileged Identity Management, access reviews

5 MICROSOFT ENTRA DOMAIN SERVICES

Microsoft Entra Domain Services provides managed domain services such as domain join, LDAP, Kerberos/NTLM authentication, and group policy without managing domain controllers yourself.

Use case: lift-and-shift apps that need traditional domain features but you want Microsoft-managed domain services.

6 KEY ADMIN DECISIONS

- Use Entra ID for Azure access, Microsoft 365 and modern cloud apps.
- Use AD DS when you need classic domain controllers, GPO and legacy authentication.
- Use Entra Domain Services when apps need domain features but you prefer managed service.
- Use P1/P2 features when you need stronger identity governance and security.
- Exam focus: know differences between Entra ID, AD DS and Entra Domain Services.

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CREATE, CONFIGURE AND MANAGE IDENTITIES

Users, groups, licenses, device registration, custom attributes and SSPR.

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1 CREATE & MANAGE USERS

- Create user with username, display name and contact info.
- Set usage location before license assignment.
- Manage profile, roles, groups and authentication methods.
- Restore or permanently remove deleted users when needed.

2 GROUPS IN MICROSOFT ENTR

- Security groups: access control and RBAC assignments.
- Microsoft 365 groups: collaboration across Teams, SharePoint, Outlook.
- Use dynamic groups when membership follows rules.
- Assign access and licenses to groups where possible.

3 LICENSE MANAGEMENT

- Assign licenses directly to users for simple labs.
- Use group-based licensing for scale.
- Review service plans inside each license.
- Troubleshoot license conflicts and location issues.

4 STEP-BY-STEP: USER LICENSE ASSIGNMENT

1. Open Entra/Admin Center → 2. Select Users → 3. Set usage location → 4. Licenses & apps → 5. Assign license → 6. Verify services

Practice exercises: assign licenses to users, change user license assignments, create groups, add users to groups, and change group license assignments.

5 DEVICE REGISTRATION, CUSTOM SECURITY ATTRIBUTES AND

- Device registration lets users register devices with Entra ID for identity-based access scenarios.
- Custom security attributes store business-specific metadata for users or service principals.
- Automatic user creation can help create identities from external or connected systems.
- Always document identity source, license model and join/registration strategy.

6 SELF-SERVICE PASSWORD RESET

- Enable SSPR for selected users or groups first.
- Choose authentication methods.
- Test password reset experience.
- Customize directory/company branding.
- Monitor registration and reset activity.

7 SSPR IMPLEMENTATION FLOW

Enable SSPR → Choose users or group → Select auth methods → Customize branding → Test reset

Exam focus: users, groups, licenses, external users, SSPR and device registration are core identity operations for Azure administrators.

AZURE RBAC, POLICY INITIATIVES AND GOVERNANCE

Secure access, enforce standards, and govern Azure resources at scale.

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1 WHAT IS AZURE RBAC?

- Authorization system for Azure resources.
- Answers: who can do what at which scope.
- Built-in roles: Owner, Contributor, Reader, User Access Administrator.
- Scopes: management group, subscription, resource group, resource.
- RBAC grants permissions; it does not enforce resource configuration.

2 RBAC ACCESS ASSIGNMENT FLOW

Select principal → Choose role → Select scope → Assign access → Verify activity log

Best practice: assign access to groups, not individuals, and use the lowest possible scope.

3 AZURE POLICY AND INITIATIVES

- Policy definition: rule that describes what is allowed or required.
- Initiative: group of policy definitions.
- Assignment: applies policy or initiative to a scope.
- Compliance: shows whether resources follow assigned policies.
- Evaluation happens at create/update and periodically.

Example policy	Purpose
Allowed locations	Restrict resource regions
Required tags	Force owner/cost tags
Allowed VM SKUs	Control size and cost

4 CLOUD ADOPTION FRAMEWORK + DESIGN PRIN

- Define governance standards before large deployments.
- Use management groups for scale.
- Apply naming, tagging, security and cost policies.
- Use initiatives for common compliance bundles.
- Automate governance with policy as code / IaC.

5 POLICY VS RBAC - REMEMBER THIS

Area	Azure RBAC	Azure Policy
Purpose	Control who can do what	Control what can be created or configured
Focus	People and permissions	Resources and compliance
Example	Contributor can create VM	Policy allows only approved VM SKUs
Exam phrase	Access assignment	Governance / compliance

6 GOVERNANCE CHECKLIST

- List access using Azure RBAC and portal.
- Grant access at the correct scope.
- View activity logs for RBAC changes.
- Create Azure Policy definitions and initiatives.
- Evaluate compliance and remediate non-compliant resources.

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CONFIGURE VIRTUAL NETWORKS IP ADDRESSING AND SUBNETS

Plan, create and connect VNets like an Azure administrator.

1 PLAN VIRTUAL NETWORKS

- A VNet is a private network in Azure.
- Plan address spaces with non-overlapping CIDR ranges.
- Create subnets for tiers: web, app, data, management.
- Reserve enough IP space for growth.
- Use separate subnets for Bastion, Gateway or Azure Firewall when needed.

VNet 10.0.0.0/16

Subnet A	Subnet B
10.0.1.0/24	10.0.2.0/24

2 CREATE VNETS AND SUBNETS

Plan CIDR → Create VNet → Create Subnets → Attach Resources → Validate IP use

- Portal path: Virtual networks > Create.
- Choose region, address space and subnet.
- Associate NSG and route table where required.
- Review + create, then deploy VM or service into subnet.

3 PUBLIC IP ADDRESSING

- Public IP gives internet connectivity to supported resources.
- Standard SKU is recommended for production/high availability.
- Public IP can be static or dynamic depending on SKU and service.
- Associate to VM NIC, Load Balancer, Application Gateway or VPN Gateway.

4 PRIVATE IP ADDRESSING

- Private IP is used inside a VNet/subnet.
- Dynamic private IP is assigned by Azure from subnet range.
- Static private IP is useful for servers, appliances and DNS references.
- Avoid IP overlap when connecting VNets or on-prem networks.

5 VIRTUAL NETWORK PEERING

VNet peering connects two VNets privately over the Azure backbone. It can connect VNets in the same region or different regions. Peering is not transitive by default.

VNet A 10.1.0.0/16 → Peering private traffic → VNet B 10.2.0.0/16

- Options include Allow VNet access, forwarded traffic and gateway transit.
- Use hub-spoke network design for shared services.

6 NETWORKING CHECKPOINT

- Can you design a VNet CIDR and subnet plan?
- Can you create public/private IPs and attach them correctly?
- Can you configure VNet peering and explain gateway transit?
- Can you troubleshoot a simple connectivity failure?

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NETWORK SECURITY, DNS, ROUTES ENDPOINTS AND PEERING

NSGs, ASGs, Azure DNS, custom routes, NVA, service endpoints and private endpoints.

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1 NETWORK SECURITY GROUPS

- NSGs filter inbound and outbound traffic at subnet or NIC level.
- Rules include priority, source, destination, port, protocol and action.
- Lower priority number is processed first.
- NSGs are stateful: return traffic is automatically allowed.
- Use effective security rules to troubleshoot.

Priority	Name	Dir	Port	Action
100	Allow-SSH	In	22	Allow
200	Deny-All-In	In	*	Deny
300	Allow-Web	In	80/443	Allow

2 APPLICATION SECURITY GROUPS

- ASGs group VMs by application role.
- Use ASGs in NSG rules instead of fixed IP addresses.
- Example: WebASG can reach AppASG on port 443.
- Helpful in multi-tier applications.

3 AZURE DNS AND DOMAIN HOSTING

- Azure DNS hosts DNS zones and records.
- Create DNS zone, add A/CNAME records, update registrar name servers.
- Alias records can point to Azure resources.
- Private DNS zones resolve names inside linked VNets.
- Use DNS checks when troubleshooting name resolution.

4 ROUTES, UDR AND NVA

- Azure creates system routes automatically.
- User-defined routes override system routes.
- Route tables are associated to subnets.
- Next hop examples: Internet, Virtual Appliance, VNet Peering, Virtual Network Gateway, None.
- NVA is a network virtual appliance such as firewall/router VM.

5 SERVICE ENDPOINTS VS PRIVATE ENDPOINTS

Feature	Service Endpoint	Private Endpoint
Purpose	Extend VNet identity to PaaS	Private IP for PaaS inside VNet
Traffic	Azure backbone	Private Link over Azure backbone
DNS impact	Usually public endpoint DNS	Requires private DNS planning
Security	Restrict service to subnet	Disable public access where possible

6 LABS TO PRACTICE

- Implement virtual networking with VNets, subnets and NSG rules.
- Create Azure DNS zone and A record; create alias record.
- Create custom route table and route traffic through NVA.
- Implement VNet peering and test connectivity.

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LOAD BALANCER, APPLICATION GATEWAY AND NETWORK WATCHER

Distribute traffic and troubleshoot connectivity with confidence.

1 AZURE LOAD BALANCER

- Layer 4 load balancer for TCP/UDP.
- Can be public or internal.
- Uses frontend IP, backend pool, health probe and load-balancing rules.
- Good for VM-based workloads and high availability.

Client → Frontend IP → Load Balancer → Backend Pool

2 APPLICATION GATEWAY

- Layer 7 HTTP/HTTPS load balancer.
- Supports URL path-based routing, host-based routing and SSL/TLS offload.
- Can use Web Application Firewall (WAF).
- Best for web applications needing advanced routing.

Users → App Gateway WAF → HTTP/HTTPS Rules → Backend Pool

3 NETWORK WATCHER

- Regional service for monitoring and troubleshooting networks.
- IP Flow Verify: check allowed/denied traffic.
- Next Hop: identify route decision.
- NSG Flow Logs: capture traffic through NSGs.
- Packet Capture: capture packets for diagnosis.
- Topology: visualize network layout.

4 CONNECTION MONITOR

- Monitor end-to-end network connectivity.
- Checks reachability, latency and packet loss.
- Can test from source to destination across Azure/on-prem.
- Useful for detecting intermittent issues and service path problems.

Source → Test Connectivity → Destination

5 CONNECTIVITY TROUBLESHOOTING FLOW

1. Can the source resolve the name? Check Azure DNS / Private DNS.
2. Can the source reach the destination IP? Use Connection Monitor or IP Flow Verify.
3. Is routing correct? Check UDR, system routes and Next Hop.
4. Are NSG or firewall rules allowing traffic? Check effective rules and logs.
5. Is the backend service healthy and listening? Check probes, ports and health.
6. Still failing? Capture packets and review topology.

6 EXAM FOCUS

- Load Balancer = Layer 4 TCP/UDP. Application Gateway = Layer 7 HTTP/HTTPS.
- Network Watcher helps troubleshoot IP flow, routes, packet capture, topology and connection monitoring.
- Know when to use public vs internal load balancer and how health probes affect traffic.

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CONFIGURE STORAGE ACCOUNTS SERVICES, ACCESS AND REDUNDANCY

Storage account types, replication, access methods and endpoint security.

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1 IMPLEMENT AZURE STORAGE

- Storage account is the top-level container for Azure Storage data.
- Choose account type based on workload, performance and cost.
- Storage services include Blob, Files, Queues and Tables.
- Use redundancy to protect data durability.

2 AZURE STORAGE SERVICES

- Blob Storage: unstructured object data.
- Azure Files: managed file shares over SMB/NFS.
- Queues: asynchronous messages between components.
- Tables: NoSQL key-value data.
- Disks: persistent managed disks for VMs.

3 ACCOUNT TYPES AND USE CASES

Type	Use case
Standard GPv2	Most general storage scenarios
Premium block blobs	High transaction blob workloads
Premium file shares	Performance-sensitive file shares
Blob account	Legacy/specialized blob-only scenarios

4 REPLICATION STRATEGIES

Option	Protection
LRS	3 copies in one datacenter
ZRS	3 copies across availability zones
GRS	LRS in primary + secondary paired region
RA-GRS	GRS + read access to secondary
GZRS	ZRS in primary + geo-replication

5 ACCESS STORAGE AND SECURE ENDPOINTS

Access methods

- Azure Portal
- Azure Storage Explorer
- AzCopy
- Azure CLI / PowerShell
- SDKs and APIs

Security controls

- Access keys rotation
- Shared Access Signatures (SAS)
- Stored access policies
- Microsoft Entra ID access
- Firewalls and VNet rules
- Private endpoints
- Encryption at rest and in transit

6 STORAGE DECISION GUIDE

Need	Recommended service
Large unstructured files, images, logs	Blob Storage
Shared file path for Windows/Linux apps	Azure Files
Asynchronous work queue	Queue Storage
Simple key-value NoSQL data	Table Storage
VM OS/data disk	Managed Disk

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BLOB STORAGE, STORAGE SECURITY AZURE FILES AND TOOLS

Containers, tiers, lifecycle, SAS, encryption, snapshots and Storage Explorer.

1 CONFIGURE AZURE BLOB STORAGE

- Create containers to organize blobs.
- Choose access tier: Hot, Cool, Cold/Archive based on access frequency.
- Lifecycle management moves/deletes blobs automatically.
- Object replication copies blobs across storage accounts.
- Manage blobs using portal, Storage Explorer, AzCopy or scripts.

Hot | Cool | Archive

2 CONFIGURE STORAGE SECURITY

- Review storage security strategy before exposing data.
- SAS gives time-bound delegated access.
- SAS URI contains signed parameters such as permissions and expiry.
- Encryption is enabled at rest by default.
- Customer-managed keys can use Azure Key Vault.
- Apply firewalls, private endpoints and least privilege.

3 CONFIGURE AZURE FILES

- Azure file shares provide SMB/NFS file shares.
- Create snapshots for point-in-time recovery.
- Soft delete helps recover deleted file shares.
- Azure File Sync can cache Azure file shares on Windows Server.
- Use Storage Explorer for easy management.

4 FILES VS BLOBS

Question	Use Azure Files	Use Blob
Need SMB/NFS share?	Yes	No
Object data / web content?	No	Yes
Lift-and-shift file server?	Yes	Sometimes
Mass unstructured objects?	Maybe	Best

5 STORAGE PRACTICAL LAB FLOW

Create storage account → Create blob container → Upload website content → Configure SAS or access → Test access and logs

Practice exercises: provide storage for a public website, manage Azure Storage security, create SAS, use Azure Storage Explorer, create Azure Files share and snapshot.

6 EXAM FOCUS POINTS

- Know redundancy choices and when to use each one.
- Know SAS, access keys, stored access policies and encryption.
- Know blob tiers, lifecycle management, versioning and soft delete.
- Know Azure Files snapshots, soft delete and File Sync basics.

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AZURE VIRTUAL MACHINES, AVAILABILITY AND VM SCALE SETS

Create, size, protect, scale and back up Azure virtual machines.

1 INTRODUCTION TO AZURE VIRTUAL MACHINE

- VMs provide IaaS compute with full OS control.
- Before creating a VM choose image, size, disks, network, authentication and region.
- VMs can be managed by Portal, CLI, PowerShell, ARM/Bicep and extensions.
- Use managed disks, NSGs, backups and monitoring.

2 CHECKLIST FOR CREATING A VM

- Resource group and region selected.
- Image selected: Windows or Linux.
- Size selected based on CPU, memory and workload.
- OS disk and data disks configured.
- VNet/subnet/NIC/NSG configured.
- Authentication and management options reviewed.

3 VM AVAILABILITY OPTIONS

Option	Purpose
Availability set	Protect from rack/fault domain failures
Availability zone	Protect from datacenter failure in a region
VM Scale Set	Scale identical VMs with load balancing
Backup	Recover VM or files after issue

4 UPDATE DOMAINS, FAULT DOMAINS AND SCALIN

- Fault domain: different hardware/rack.
- Update domain: planned maintenance grouping.
- Vertical scaling: change VM size.
- Horizontal scaling: add more instances.
- Autoscale uses metrics, schedules or rules.

5 IMPLEMENT VIRTUAL MACHINE SCALE SETS

VM Scale Sets deploy and manage a group of load-balanced VM instances. Use VMSS for scalable, highly available, identical compute nodes.

Create VMSS → Select image & size → Set scaling rules → Configure LB & health probes → Monitor and upgrade

- Practice: create VMSS, configure autoscale, compare vertical and horizontal scaling.

6 BACK UP YOUR VIRTUAL MACHINES

- Enable Azure Backup for important VMs.
- Define backup policy, schedule and retention.
- Test restore: full VM, disk or file-level recovery.

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APP SERVICE, CONTAINERS AND ARM/BICEP DEPLOYMENTS

PaaS web apps, deployment slots, containers and infrastructure as code.

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1 AZURE APP SERVICE PLANS

- App Service Plan defines compute resources and pricing tier.
- Scale up changes plan size/tier.
- Scale out increases instance count.
- Autoscale can respond to metrics and schedules.
- Choose plan based on features, cost and scaling needs.

2 CONFIGURE AZURE APP SERVICE

- Create app with App Service.
- Configure CI/CD and deployment options.
- Use deployment slots such as dev, stage and production.
- Secure app with HTTPS, identity and access restrictions.
- Add custom domain and certificate.
- Back up and restore app; use Application Insights.

3 AZURE CONTAINER INSTANCES

- Containers package app and dependencies.
- ACI runs containers without managing servers.
- Container groups share lifecycle, network and storage.
- Good for short jobs, tests and simple services.
- Compare containers to VMs: containers start faster and are lighter.

4 AZURE CONTAINER APPS

- Serverless container platform for microservices and event-driven apps.
- Supports revisions, scaling and traffic splitting.
- Good when you need containers without managing Kubernetes.
- Use with container images from registries.

5 ARM TEMPLATES AND BICEP FOR COMPUTE DEPLOYMENT

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Azure Resource Manager templates and Bicep allow repeatable, version-controlled infrastructure deployments. You should be able to interpret, modify, deploy and export templates.

Exam focus: know App Service plans, scale up/out, deployment slots, custom domains/TLS, containers, ACI/ACA and ARM/Bicep deployment flow.

6 CHOOSE THE RIGHT COMPUTE OPTION

Need	Best fit
Full OS control or legacy app	Virtual Machine
Identical scalable VM fleet	VM Scale Set
Web app/API with managed platform	App Service
Portable modern app or job	Containers / ACI / ACA

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AZURE MONITOR, LOG ANALYTICS ALERTS, BACKUP AND SITE RECOVERY

Monitor resources, query logs, configure alerts, protect VMs and plan recovery.

1 AZURE MONITOR FOR VMS

- Azure Monitor collects metrics and logs.
- Metrics are numeric time-series data such as CPU or disk.
- Logs contain detailed records, events and traces.
- Use Metrics Explorer, dashboards and VM insights.
- Collect guest data with VM insights and agents when required.

2 LOG ANALYTICS WORKSPACE AND KQL

A Log Analytics workspace stores log data. KQL helps search, filter and summarize logs.

```
AzureActivity
| where TimeGenerated > ago(24h)
| summarize count() by OperationName
```

Practice simple filters, summarize, time ranges and searching activity logs.

3 ALERTS: FROM DETECTION TO ACTION

Metric or Log → **Alert Rule** → **Action Group** → **Notify or Act**

- Alert rule defines condition.
- Action group defines notifications/actions such as email, SMS, webhook, ITSM, Logic App.
- Alert processing rules can suppress, route or modify alert handling.

4 AZURE BACKUP

- Azure Backup protects VMs, files, folders, SQL and more.
- Use Backup vault or Recovery Services vault based on scenario.
- Create backup policy: schedule, retention and frequency.
- Run backup and test restore operations.
- Review backup reports and alerts.

5 PROTECT VIRTUAL MACHINES BY USING AZURE BACKUP

VM Resource → **Backup Policy** → **Backup Vault** → **Backup Data** → **Restore VM or Files**

- Exercise: back up an Azure virtual machine.
- Exercise: restore Azure virtual machine data.
- Restore options include original location, alternate location, disk restore or file-level recovery.

6 AZURE SITE RECOVERY - DISASTER RECOVERY FLOW

Primary Region → **Continuous Replication** → **Secondary Region** → **Failover During Outage** → **Failback When Ready**

Exam focus: create vaults, backup policies, backup/restore operations, ASR replication, failover and alerts/reports.

HANDS-ON ROADMAP, EXAM TIPS FINAL CHECKLIST AND QUICK REVISION

Practice the modules, revise core services, and become exam-ready.

1 30-DAY HANDS-ON ROADMAP

Week 1
Cloud Shell, ARM templates, Azure core architecture, Entra ID, users/groups, RBAC

Week 2
Azure Policy, governance, VNets, subnets, NSGs, DNS and peering

Week 3
Routes, NVA, Load Balancer, App Gateway, Network Watcher, Storage accounts

Week 4
Blob, Files, VMs, VMSS, App Service, Containers, Monitor, Backup, ASR and revision

2 LAB IDEAS CHECKLIST

- Start Cloud Shell and run basic Azure CLI commands
- Deploy a resource group using ARM template
- Create users, groups, licenses and SSPR
- Assign RBAC and apply Azure Policy
- Build VNet, subnet, NSG, peering and DNS zone
- Create storage account, Blob container and file share
- Deploy VM, VMSS, App Service and container
- Configure alerts, VM backup and restore test

3 EXAM TIPS

- Read every scenario carefully before choosing the service.
- Know scopes: management group > subscription > resource group > resource.
- Compare similar services: Load Balancer vs App Gateway, Service Endpoint vs Private Endpoint, Blob vs Files.
- Practice portal, CLI, PowerShell and ARM/Bicep.
- Review weak areas with hands-on labs, not only theory.

4 QUICK REVISION - 5 SKILL DOMAINS

Domain	Weight
Identities & Governance	20-25%
Storage	15-20%
Compute	20-25%
Virtual Networking	15-20%
Monitor & Maintain	10-15%

5 FINAL READINESS CHECKLIST

- I can explain Azure hierarchy, regions, availability zones and resource groups.
- I can manage Entra users, groups, licenses, device registration and SSPR.
- I can assign RBAC roles and understand scope inheritance.
- I can use Azure Policy, initiatives, tags, locks, budgets and Advisor.
- I can configure VNets, subnets, IPs, NSGs, DNS, peering, routes, endpoints and load balancing.
- I can create and secure storage accounts, Blob containers and Azure file shares.
- I can deploy VMs, availability, VMSS, App Service, containers and ARM/Bicep.
- I can configure Azure Monitor, Log Analytics, alerts, Backup and Site Recovery.

6 FINAL MESSAGE

Practice in Azure, revise the core services, and stay consistent. AZ-104 is about real administrator skills: secure, govern, deploy, connect, monitor and recover Azure resources.